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Electrical assembly

Abstract

An assembly includes a module including a housing having a first portion, a second portion including a projection, and a first interior cavity; and a circuit board disposed at least partially in the first interior cavity; and an additional module connected under tension with the module, the additional module including: an additional housing having a recess and an additional interior cavity; and an additional circuit board disposed at least partially in the additional interior cavity. The second portion of the housing may be disposed at least partially in the additional interior cavity. The projection of the housing may be disposed at least partially in the recess of the additional housing. The circuit board may be electrically connected to the additional circuit board.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure generally relates to electrical assemblies, including modules that may, for example, be utilized in connection with and/or incorporated into vehicles.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) While the claims are not limited to a specific illustration, an appreciation of various aspects may be gained through a discussion of various examples. The drawings are not necessarily to scale, and certain features may be exaggerated or hidden to better illustrate and explain an innovative aspect of an example. Further, the exemplary illustrations described herein are not exhaustive or otherwise limiting, and embodiments are not restricted to the precise form and configuration shown in the drawings or disclosed in the following detailed description. Exemplary illustrations are described in detail by referring to the drawings as follows:

(2) FIGS. **1** and **2** are perspective views generally illustrating an embodiment of an electrical assembly according to teachings of the present disclosure.

(3) FIGS. **3A**, **3B**, and **4** are perspective views general illustrating an embodiment of a first module according to teachings of the present disclosure.

(4) FIGS. **5** and **6** are perspective views general illustrating an embodiment of a second module according to teachings of the present disclosure.

(5) FIGS. **7** and **8** are perspective views general illustrating an embodiment of a third module according to teachings of the present disclosure.

(6) FIG. **9** is a cross-sectional view generally illustrating an embodiment of an electrical assembly according to teachings of the present disclosure.

(7) FIG. **10** is a top view generally illustrating an embodiment of an electrical assembly according to teachings of the present disclosure.

(8) FIG. **11** is a partial top view generally illustrating an embodiment of an electrical assembly according to teachings of the present disclosure.

(9) FIG. **12** is a partial perspective view generally illustrating an embodiment of an electrical assembly according to teachings of the present disclosure.

(10) FIG. **13** is a partial cross-sectional view generally illustrating an embodiment of an electrical assembly according to teachings of the present disclosure.

(11) FIG. **14** is a perspective view generally illustrating an embodiment of a method of assembling an electrical assembly according to teachings of the present disclosure.

DETAILED DESCRIPTION

(12) Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the various described embodiments.

However, it will be apparent to one of ordinary skill in the art that the various described embodiments may be practiced without these specific details. In other instances, well-known methods, procedures, components, circuits, and networks have not been described in detail so as not to unnecessarily obscure aspects of the embodiments.

(13) FIGS. **1** and **2** present an electrical assembly **20** including a first module **30**, a second module **40**, and/or a plurality of third modules **50** (e.g., third modules **50A-50I**). The first module **30** is configured for connection with a third module **50**. The second module **40** is configured for connection with a third module **50**. A third module **50** is configured for connection with the first module **30**, the second module **40**, and/or one or more additional third modules **50**. The first module **30**, the second module **40**, and/or the third module(s) **50** may be electrically connected. In some example configurations, the third module(s) **50** are disposed at least partially between the first module **30** and the second module **40**. For example, the third module(s) **50** may be intermediate or additional modules, the first module **30** may be a first end module, and the second module **40** may be a second end module. Some or all of modules **30**, **40**, **50** (e.g., at least adjacent modules) are connected under tension. For example and without limitation, the first module **30** may be connected under tension with a third module **50** (e.g., third module **50A**), the second module **40** may be connected under tension with the third module **50** and/or an additional third module **50**

(e.g., third module **50I**), and/or the third module **50** may be connected under tension with the first module **30**, the second module **40**, and/or an additional third module **50** (e.g., third module **50B**).

(14) With some embodiments, the assembly **20** may be used in connection with one or more bladders of a vehicle seat **22** of a vehicle **24**. For instance, a module (e.g., modules **30**, **40**, **50**) may include a bladder control module for a vehicle seat **22**. A vehicle seat **22** may include the electrical assembly **20** and one or more fluid bladders **26** (e.g., fluid bladders/groups **26A-26I**) in fluid communication with the electrical assembly **20**. One or more of the modules **30**, **50** may be configured to control fluid flow to and/or from a respective bladder **26** or group of bladders **26**, such as to and/or from a fluid source **28** (e.g., a pump, tank, etc.). For example, one or more modules **30**, **50** may include a fluid valve **68**, **108** that may be controlled via electronics of the module **30**, **50**, such as via a circuit board **62**, **102**, a solenoid, expanding/contracting wires, and/or other electronics. The valves **68**, **108** of adjacent modules **30**, **50** are in fluid communication. For example, connecting adjacent modules **30**, **50** may include connecting valves **68**, **108** for fluid communication. The first module **30** may include an inlet port **76** that may be connected to the fluid source **28**. The inlet port **76** may be in fluid communication with the fluid valve **68**.

(15) Examples of an electrical assembly **20** are shown including nine third modules **50**, however the assembly **20** may include more or less than nine third modules **50** (e.g., an electrical assembly **20** may have a modular configuration). For example, some configurations may include one third module **50** and other configurations may include, two, three, four, five, or other numbers of third modules **50**. The number of third modules **50** may correspond to the number of components (e.g., bladders, fluid actuators, etc.) to be controlled.

(16) FIGS. **3A**, **3B**, and **4** illustrate a first module **30** including a housing **60** and a circuit board **62**. The housing **60** includes an interior cavity **64** and the circuit board **62** is disposed at least partially in the interior cavity **64**. The housing **60** includes one or a plurality of projections **70** (e.g., projections **70A-70D**) and/or one or a plurality of ramped portions **72** (e.g., ramped portions **72A-72F**). A first projection **70A** and a second projection **70B** may extend from a first side wall **66A** of the housing **60** (e.g., an upper wall) and/or a third projection **70C** and a fourth projection **70D** may extend from a second sidewall **66B** of the housing **60** (e.g., a lower wall). A projection **70** may include a ramped configuration. The housing **60** is shown including four projections **70**, however the housing **60** may include more or less than four projections **70**. In some implementations, one or more ramped portions (e.g., ramped portions **72A-72C**) extend from the first sidewall **66A** of the housing **60**, and/or one or more ramped portions (e.g., ramped portions **72D-72F**) extend from the second sidewall **66B** of the housing **60**. A ramped portion **72** is configured to provide an interference fit with an adjacent module (e.g., a third module **50**). The housing **60** is shown including six ramped portions **72**, however the housing **60** may include more or less than six ramped portions **72**. The housing **60** may comprise one or more materials that are configured to deform (e.g., stretch, bend, flex, etc.), at least to some degree, during assembly, such as one or more plastics and/or polymers. In some examples, the circuit board **62** includes an electrical connector **74** that connects to an electrical connector of an adjacent module (e.g., an adjacent third module **50**).

(17) FIGS. **5** and **6** illustrate a second module **40** including a housing **80** having an interior cavity **84**. In some configurations, the housing **80** includes a plurality of recesses **90** (e.g., recesses **90A-90D**). A first recess **90A** and a second recess **90B** may be disposed in a first sidewall **86A** of the housing **80** (e.g., an upper wall) and/or a third recess **90C** and a fourth recess **90D** may be disposed in a second sidewall **86B** of the housing **80** (e.g., a lower wall). The housing **80** is shown including four recesses **90**, however the housing **80** may include more or less than four recesses **90**. The housing **80** may comprise one or more materials that are configured to deform (e.g., stretch, bend, flex, etc.), at least to some degree, during assembly, such as one or more plastics and/or polymers. Optionally, the second module **40** includes a circuit board at least partially disposed in the interior cavity **84**. The circuit board may include an electrical connector that connects to an electrical

connector of an adjacent module. In other configurations, the second module **40** does not include a circuit board or other components and functions as a cover for an adjacent third module **50**.

(18) FIGS. **7** and **8** illustrate a third module **50** including a housing **100** and a circuit board **102**. The housing **100** includes an interior cavity **104** and the circuit board **102** is disposed at least partially in the interior cavity **104**. The housing **100** includes a first portion **110** and a second portion **112** offset from the first portion **110** in an assembly direction (e.g., an X-direction). The first portion **110** may be larger than the second portion **112**. In some configurations, the first portion **110** includes a plurality of recesses **120** (e.g., recesses **120A-120D**). A first recess **120A** and a second recess **120B** may be disposed in a first sidewall **122A** of the first portion **110** (e.g., an upper wall) and/or a third recess **120C** and a fourth recess **120D** may be disposed in a second sidewall **122B** of the first portion **110** (e.g., a lower wall). The housing **100** is shown including four recesses **120**, however the housing **100** may include more or less than four recesses **120**. The number of recesses **120** may correspond to the number of projections **70** of the first module **30**.

(19) In some implementations, the second portion **112** of the housing **100** includes a plurality of projections **130** (e.g., projections **130A-130D**). A first projection **130A** and a second projection **130B** may extend from a first sidewall **132A** of the second portion **112** (e.g., an upper wall) and/or a third projection **130C** and a fourth projection **130D** may extend from a second sidewall **132B** of the second portion **112** (e.g., a lower wall). A projection **130** may include a ramped configuration. The housing **100** is shown including four projections **130**, however the housing **100** may include more or less than four projections **130**. The housing **100** may comprise one or more materials that are configured to deform (e.g., stretch, bend, flex, etc.), at least to some degree, during assembly, such as one or more plastics and/or polymers. In some examples, the circuit board **102** includes at least one electrical connector **134**, **136**. An electrical connector **134**, **136** connects to an electrical connector of an adjacent module. For example, an electrical connector **136A** of a third module **50A** may connect with an electrical connector **134B** of an adjacent third module **50B**. Additionally or alternatively, an electrical connector **134A** may connect with an electrical connector **74** of the first module **30**. In some configurations, the electrical connectors **134**, **136** may provide power and/or data/instructions from the first module **30** to the one or more third modules **50**.

(20) With reference to FIG. **9**, a first module **30** and two third modules **50A-B** are shown. In some configurations, a housing **100** of a third module **50** includes a first sidewall **132A** and/or a second sidewall **132B** that are tapered, such as to provide an interference fit with an adjacent module. For example, a Z-dimension of the second portion **112** of the housing **100** may increase in the X-direction, such as toward the first module **30**. In some implementations, the first sidewall **132A** and/or the second sidewall **132B** may include tapered edges to facilitate insertion of an adjacent module **30**, **50**.

(21) Referring now to FIG. **10**, a first sidewall **122A** and/or a second sidewall **122B** of a housing **100** includes a recessed portion **140** (e.g., recessed portion **140A**, **140B**) such that a gap/clearance is disposed between sections of adjacent modules. For example and without limitation, a third module **50A** may be connected with an additional third module **50B** such that a recessed portion **140A** is disposed between a first sidewall **122A** of the third module **50A** and a first sidewall **122A'** of the additional third module **50B**. Additionally or alternatively, a recessed portion **140B** is disposed between a second sidewall **122B** of the third module **50A** and a second sidewall **122B'** of the additional third module **50B**. The recessed portions **140A-B** may result in reduced contact in the X-direction between adjacent sidewalls **122A**, **122A'** and **122B**, **122B'**. For example, the recessed portions **140** may extend along at least 70% of the sidewalls **122A**, **122A'** and contact in the X-direction between sidewalls **122A**, **122A'** may be along 30% or less of their Y-dimension lengths. A depth of the recessed portions **140** in the X-direction may correspond to an offset **170** between adjacent modules described below. For example, the depth of the recessed portions **140** may be at least as great as the offset **170**, which may facilitate connecting adjacent modules (e.g., third modules **50A**, **50B**), such as by providing room for stretching of the sidewalls.

(22) With reference to FIGS. 7 and 12, a housing (e.g., housings 60, 80, 100) may include at least one track 150 and/or at least one channel 160. For example and without limitation, a second portion 112 of a housing 100 may include at least one track 150 and a first portion 110 of the housing 100 may include at least one channel 160. In some implementations, a third sidewall 132C of the second portion 112 may include a first track 150A and a fourth sidewall 132D of the second portion 112 may include a second track 150B. A third sidewall 122C of the first portion 110 may include a first channel 160A and a fourth sidewall 122D of the first portion 110 may include a second channel 160B. A track 150 may include a dovetail-shaped configuration. A channel 160 may include a shape that corresponds to a shape of a track 150. In an assembled configuration, a module (e.g., third module 50A) may be connected with an adjacent module (e.g., third module 50B) such that a track 150 of the module is disposed in and/or engaged with a channel 160 of the adjacent module (see, e.g., FIG. 12). Engagement between a track 150 of a module and channel 160 of an adjacent module may restrict and/or prevent relative movement between the two modules in the Y-direction and the Z-direction, and may not substantially restrict movement in the X-direction.

(23) With reference to FIGS. 1, 2, 9, and 10, in an assembled configuration, adjacent modules are connected under tension. For example and without limitation, the first module 30 may be connected under tension with a third module 50 (e.g., third module 50A), the second module 40 may be connected under tension with the third module 50A and/or an additional third module 50 (e.g., third module 50I), and/or the third module 50 may be connected under tension with the first module 30, the second module 40, and/or an additional third module 50 (e.g., third module 50B).

(24) In some implementations, the first module 30 is connected with a third module 50A such that a portion of the first module 30 is disposed at least partially in an interior cavity 104A of the third module 50A, a projection 70 of the first module 30 is disposed at least partially within a recess 120 of the third module 50A, and/or a circuit board 62 of the first module 30 is electrically connected with a circuit board 102A of the third module 50A via electrical connectors 74, 134.

(25) In some instances, the third module 50A may be connected with an additional third module 50B such that a second portion 112A of a housing 100A of the third module 50A is at least partially disposed in an interior cavity 104B of the additional third module 50B. A projection 130 of the third module 50A is at least partially disposed within a recess 120' of the additional third module 50B. The electrical connector 134B of the third module 50B is connected (e.g., directly) to the electrical connector 134A of the third module 50 such that the circuit board 102A of the third module 50A is electrically connected with a circuit board 102B of the additional third module 50B (see, e.g., FIG. 13).

(26) Referring now to FIG. 11, the third module 50A may be connected under tension with the additional third module 50B at least in part because of a pre-assembly/flex offset 170 between a projection 130 of the third module 50A and a surface 180 defining a recess 120' of the additional third module 50B. The offset 170 may be the X-direction distance between the projection(s) 70, 130 of one module 30, 50 and the recess(es) 90, 120 of the adjacent module 40, 50 when the one module is fully inserted into the adjacent module. For example, when one module 30, 50 is fully inserted into the adjacent module 40, 50, the projection(s) 70, 130 may not be aligned with the recess(es) 90, 120. The housing 80, 100 of the adjacent module 40, 50 may then be stretched at least the distance of the offset such that projection(s) and the recess(es) 90, 120 are aligned for engagement. The pre-assembly/flex offset may, for example and without limitation, be approximately 0.2 mm to 0.3 mm.

(27) In some examples, the second module 40 may be connected with a third module 50 such that a second portion 112 of a housing 100 of the third module 50 is at least partially disposed within an interior cavity 84 of the second module 40. A projection 130 of the third module 50 is at least partially disposed within a recess 90 of the second module 40. If the second module 40 includes a circuit board, the circuit board may be electrically connected with a circuit board 102 of the third module 50.

(28) Adjacent modules **30**, **50** are connected under tension such that electrical connectors **74**, **134** of the adjacent modules **30**, **50** are connected under tension which restricts movement between the electrical connectors **74**, **134**, so that the adjacent modules **30**, **50** do not become unintentionally disconnected and wear/fretting of the electrical connectors **74**, **134** is limited. For example and without limitation, as shown in FIG. **13**, a third module **50A** may be connected with an additional third module **50B** such that an electrical connector **134A** of the third module **50A** is connected under tension with an additional electrical connector **134B** of the additional module **50B** and/or a circuit board **102A** of the third module **50A** is electrically connected with an additional circuit board **102B** of the additional third module **50B**.

(29) In one example assembled configuration, a first module **30** (e.g., a first end module) is disposed partially in a third module **50A** (e.g., a first additional module), the third module **50A** is disposed partially in another third module **50B** (e.g., a second additional module), and/or the another third module **50B** is partially disposed in yet another third module **50C** (e.g., a third additional module) or the second module **40** (e.g., a second end module).

(30) A method of assembling an electrical assembly **20** includes connecting at least two modules **30**, **40**, **50** together. In configurations with one third/intermediate module **50**, the third module **50** is connected with a first module **30** (e.g., a first end module) and a second module **40** (e.g., a second end module). Connecting the third module **50** with the first module **30** includes inserting a portion of the first end module **30** into the first portion of the housing **100** of the third module. Insertion may continue until the housing **100** contacts the projections **70** and/or the ramped portions **72** of the housing **60** of the first end module **30**. An offset **170** between the projections **70** of the housing **60** and the recesses **120** of the housing **100** may result in the projections **70** not initially engaging the recesses **120**. The housing **100** may then be stretched in the X-direction (e.g., to compensate for the offset **170**) such that the recesses **120** extend at least to the end of the projections **70** and the projections **70** can be inserted into/engaged with the recesses **120**. Once the projections **70** are inserted into/engaged with the recesses **120**, the housing **100** may be released from being stretched and may move/retract toward its original position (e.g., due to a resiliency of the material(s) of the housing **100**), but may not move/retract completely its original position, which may result in the housings **60**, **100** being connected under tension of the still partially stretched housing **100**. This tension biases the housings **60**, **100** toward each other and limits and/or substantially prevents relative movement therebetween. During insertion and/or stretching of the housing **100** in the X-direction, the housing **100** may also be stretched and/or deformed in the Z-direction. For example, the outer dimensions of the housing **60** (e.g., proximate the projections **70** and/or ramped portions **72**) may be similar to or even at least slightly larger than the inner dimensions of the first portion **110** of the housing **100** (e.g., in the Z-direction) such that the first portion **110** deflects outward at least to some degree, which may cause the first portion **110** to apply a Z-direction force to the housing **60** (e.g., resulting from the resiliency of material(s) of the housing **100**). This deflection and force may result in the housings **60**, **100** be connected under tension in two directions, the X-direction and the Z-direction. The ramped portions **72** may enhance/promote deflection of the first portion **110** in the Z-direction, which may increase the tension between adjacent modules.

(31) In configurations with a single third/intermediate module **50**, the second module **40** is connected to the opposite side of the third module **50** from the first module. The connection process may be similar to the connection of the third module **50** with the first module **30**. For example, the second portion **112** of the third module **50** may be inserted into the interior cavity **84** of the second module **40** and the housing **80** may be stretched in the X-direction and/or the Z-direction to allow the projections **130** of the third module **50** to be inserted into the recesses **90** of the second module **40**. The housing **80** may be released from stretching forces, but may not fully return to its original position/shape such that the housings **80**, **100** are connected under tension. For example, a sidewall **122A** may be stretched from an original position to a stretched position, and, once released, may move back toward its original position to an engaged position between the

original position and the stretched position.

(32) FIG. 14 illustrates a method of connecting two third/intermediate modules **50** of an electrical assembly **20** that includes a plurality of third/intermediate modules **50**. The third modules **50** may be connected to each other in a similar manner as the first module **30** and the adjacent third module **50** described above. A third module (e.g., third module **50C**) may be connected to an adjacent module (e.g., third module **50B**) such that the module is connected under tension to the adjacent module. Connecting the adjacent modules may include moving one module in an assembly direction **200** (e.g., parallel to the X-axis) such that a portion of the adjacent module (e.g., second portion **112B**) is disposed in an interior cavity (e.g., interior cavity **104C**) of the module.

Connecting two modules may include manipulating (e.g., stretching) a housing of the module (e.g., housing **100C**) in a vertical direction (e.g., parallel to the Z-axis) and a horizontal direction (e.g., parallel to the X-axis) such that a projection (e.g., projections **130A'-130D'**) of the adjacent module is disposed within a recess (e.g., recesses **120A''-120D''**) of the housing of the module and the module is connected under the tension to the adjacent module. In some instances, one or more additional recesses (e.g., **140A''** and **140B''**) of the housing helps facilitate the manipulation of the housing over the additional housing (e.g., housing **100B**) of the adjacent module. In some examples, prior to the manipulation, the housing is in a first position, during the manipulation the second housing deforms (e.g., is “stretched”) to a second position, and when released from the deformation, the second housing moves to a third position which is different than the first position.

(33) In at least some assembled configurations, each of the modules **30**, **40**, **50** may be at least indirectly connected under tension with each other module **30**, **40**, **50** of an electrical assembly. For example and without limitation, the first module **30** may be indirectly connected under tension with the second module **40** via one more third/intermediate modules **50**.

(34) In some implementations, the third module **50A** is connected with the additional third module **50B** such that a gap **190** in the Z-direction is disposed between the first and second sidewalls **132A'**, **132B'** of the housing **100B** of the additional third module **50B** and ends of the first and second sidewalls **132A**, **132B** of the housing **100A** of the third module **50A** (see, e.g., FIG. 9). The gap **190** may, for example and without limitation, be triangular.

(35) While examples are described with inserted portions of modules **30**, **50** including projections **70**, **130** received by recesses **90**, **120** of a receiving portion of a module **40**, **50**, such a configuration may be reversed and/or mixed such that inserted portions of modules include recesses or recesses and projections.

(36) This disclosure includes, without limitation, the following embodiments:

(37) 1. An assembly, comprising: a module including: a housing having a first portion, a second portion including a projection, and a first interior cavity; and a circuit board disposed at least partially in the first interior cavity; and an additional module connected under tension with the module, the additional module including: an additional housing having a recess and an additional interior cavity; and an additional circuit board disposed at least partially in the additional interior cavity; wherein the second portion of the housing is disposed at least partially in the additional interior cavity; the projection of the housing is disposed at least partially in the recess of the additional housing; and the circuit board is electrically connected to the additional circuit board.

(38) 2. The assembly according to embodiment 1, wherein the projection extends from a first side of the housing; the housing includes a second projection that extends from a second side of the housing; and the projection and the second projection include a ramped configuration.

(39) 3. The assembly according to any preceding embodiment, wherein the housing includes a ramped portion to provide an interference fit with the additional housing.

(40) 4. The assembly according to any preceding embodiment, wherein the projection is disposed on the ramped portion.

(41) 5. The assembly according to any preceding embodiment, wherein the additional housing includes an additional housing first portion having the recess and an additional housing second

portion having an additional housing projection.

(42) 6. The assembly according to any preceding embodiment, wherein the additional housing includes a first sidewall and a second sidewall; and the first sidewall and the second sidewall of the additional housing include tapered edges.

(43) 7. The assembly according to any preceding embodiment, wherein the first sidewall and the second sidewall of the additional module include an additional recess such that in an assembled configuration a gap/clearance is disposed between sections of adjacent modules.

(44) 8. The assembly according to any preceding embodiment, wherein the housing of the module includes a track; and the first portion of the additional housing includes a channel engaged with track such that relative movement of the module and the additional module is restricted in a lateral direction and a vertical direction.

(45) 9. The assembly according to any preceding embodiment, wherein the module is a first end module; the assembly further comprises a second end module; the additional module is disposed at least partially between the first end module and the second end module; the second end module includes a second end module housing having a second end module recess and a second end module interior cavity; and the second end module is connected, at least indirectly, with the additional module.

(46) 10. The assembly according to any preceding embodiment, wherein the first end module and the second end module are connected, at least indirectly, under tension with the additional module.

(47) 11. The assembly according to any preceding embodiment, wherein the additional housing includes an additional housing first portion having the recess and an additional housing second portion having an additional housing projection; and the second portion of the additional housing is at least partially disposed in the second end module interior cavity.

(48) 12. The assembly according to any preceding embodiment, including a second additional module connected under tension with one or both of the additional module and the second end module.

(49) 13. The assembly according to any preceding embodiment, wherein the second additional module and the additional module are connected under tension via a pre-assembly offset between the projection of the additional module and a second additional module recess of the second additional module.

(50) 14. The assembly according to any preceding embodiment, wherein the first end module, the additional module, and the second additional module are electrically connected.

(51) 15. The assembly according to any preceding embodiment, wherein the first end module, the additional module, and the second additional module include respective fluid valves that are in fluid communication with each other.

(52) 16. The assembly according to any preceding embodiment wherein walls of the additional module and the second additional module include recessed portions that facilitate stretching.

(53) 17. The assembly according to any preceding embodiment, wherein the additional module includes an additional module projection; the second additional module includes a second additional module recess; and a depth of the recessed portions corresponds to an offset of the additional module projection and the second additional module recess.

(54) 18. The assembly according to any preceding embodiment, wherein the second additional module includes a second additional module housing; the second additional module housing includes a first sidewall and a second sidewall; and the first sidewall and the second sidewall of the second additional module housing include tapered edges.

(55) 19. The assembly according to any preceding embodiment, wherein the first end module is disposed partially in the additional module; the additional module is disposed partially in the second additional module; and the second additional module is partially disposed in a third additional module or the second end module.

(56) 20. A vehicle seat, comprising the assembly according to any preceding embodiment; and one

or more fluid bladders in fluid communication with the assembly.

(57) Various examples/embodiments are described herein for various apparatuses, systems, and/or methods. Numerous specific details are set forth to provide a thorough understanding of the overall structure, function, manufacture, and use of the examples/embodiments as described in the specification and illustrated in the accompanying drawings. It will be understood by those skilled in the art, however, that the examples/embodiments may be practiced without such specific details. In other instances, well-known operations, components, and elements have not been described in detail so as not to obscure the examples/embodiments described in the specification. Those of ordinary skill in the art will understand that the examples/embodiments described and illustrated herein are non-limiting examples, and thus it can be appreciated that the specific structural and functional details disclosed herein may be representative and do not necessarily limit the scope of the embodiments.

(58) Reference throughout the specification to “examples,” “in examples,” “with examples,” “various embodiments,” “with embodiments,” “in embodiments,” or “an embodiment,” or the like, means that a particular feature, structure, or characteristic described in connection with the example/embodiment is included in at least one embodiment. Thus, appearances of the phrases “examples,” “in examples,” “with examples,” “in various embodiments,” “with embodiments,” “in embodiments,” or “an embodiment,” or the like, in places throughout the specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more examples/embodiments. Thus, the particular features, structures, or characteristics illustrated or described in connection with one embodiment/example may be combined, in whole or in part, with the features, structures, functions, and/or characteristics of one or more other embodiments/examples without limitation given that such combination is not illogical or non-functional. Moreover, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the scope thereof.

(59) It should be understood that references to a single element are not necessarily so limited and may include one or more of such element. Any directional references (e.g., plus, minus, upper, lower, upward, downward, left, right, leftward, rightward, top, bottom, above, below, vertical, horizontal, clockwise, and counterclockwise) are only used for identification purposes to aid the reader's understanding of the present disclosure, and do not create limitations, particularly as to the position, orientation, or use of examples/embodiments.

(60) “One or more” includes a function being performed by one element, a function being performed by more than one element, e.g., in a distributed fashion, several functions being performed by one element, several functions being performed by several elements, or any combination of the above.

(61) It will also be understood that, although the terms first, second, etc. are, in some instances, used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the various described embodiments. The first element and the second element are both element, but they are not the same element.

(62) The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations,

elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(63) Joinder references (e.g., attached, coupled, connected, and the like) are to be construed broadly and may include intermediate members between a connection of elements, relative movement between elements, direct connections, indirect connections, fixed connections, movable connections, operative connections, indirect contact, and/or direct contact. As such, joinder references do not necessarily imply that two elements are directly connected/coupled and in fixed relation to each other. Connections of electrical components, if any, may include mechanical connections, electrical connections, wired connections, and/or wireless connections, among others. Uses of “e.g.” and “such as” in the specification are to be construed broadly and are used to provide non-limiting examples of embodiments of the disclosure, and the disclosure is not limited to such examples.

(64) While processes, systems, and methods may be described herein in connection with one or more steps in a particular sequence, it should be understood that such methods may be practiced with the steps in a different order, with certain steps performed simultaneously, with additional steps, and/or with certain described steps omitted.

(65) As used herein, the term “if” is, optionally, construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” is, optionally, construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.

(66) All matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative only and not limiting. Changes in detail or structure may be made without departing from the present disclosure.

Claims

1. An assembly, comprising: a module including: a housing having a first portion, a second portion including a projection, and a first interior cavity; and a circuit board disposed at least partially in the first interior cavity; and an additional module connected under tension with the module, the additional module including: an additional housing having a recess and an additional interior cavity; and an additional circuit board disposed at least partially in the additional interior cavity; wherein the second portion of the housing is disposed at least partially in the additional interior cavity; the projection of the housing is disposed at least partially in the recess of the additional housing; and the circuit board is electrically connected to the additional circuit board.
2. The assembly of claim 1, wherein: the projection extends from a first side of the housing; the housing includes a second projection that extends from a second side of the housing; and the projection and the second projection include a ramped configuration.
3. The assembly of claim 1, wherein the housing includes a ramped portion to provide an interference fit with the additional housing.
4. The assembly of claim 3, wherein the projection is disposed on the ramped portion.
5. The assembly of claim 1, wherein the additional housing includes an additional housing first portion having the recess and an additional housing second portion having an additional housing projection.
6. The assembly of claim 1, wherein the additional housing includes a first sidewall and a second sidewall; and the first sidewall and the second sidewall of the additional housing include tapered edges.
7. The assembly of claim 6, wherein the first sidewall and the second sidewall of the additional module include an additional recess such that in an assembled configuration a gap/clearance is

disposed between sections of adjacent modules.

8. The assembly of claim 1, wherein the housing of the module includes a track; and the first portion of the additional housing includes a channel engaged with track such that relative movement of the module and the additional module is restricted in a lateral direction and a vertical direction.
 9. The assembly of claim 1, wherein the module is a first end module; the assembly further comprises a second end module; the additional module is disposed at least partially between the first end module and the second end module; the second end module includes a second end module housing having a second end module recess and a second end module interior cavity; and the second end module is connected, at least indirectly, with the additional module.
 10. The assembly of claim 9, wherein the first end module and the second end module are connected, at least indirectly, under tension with the additional module.
 11. The assembly of claim 10, wherein the additional housing includes an additional housing first portion having the recess and an additional housing second portion having an additional housing projection; and the second portion of the additional housing is at least partially disposed in the second end module interior cavity.
 12. The assembly of claim 9, including a second additional module connected under tension with one or both of the additional module and the second end module.
 13. The assembly of claim 12, wherein the second additional module and the additional module are connected under tension via a pre-assembly offset between the projection of the additional module and a second additional module recess of the second additional module.
 14. The assembly of claim 12, wherein the first end module, the additional module, and the second additional module are electrically connected.
 15. The assembly of claim 14, wherein the first end module, the additional module, and the second additional module include respective fluid valves that are in fluid communication with each other.
 16. The assembly of claim 12, wherein walls of the additional module and the second additional module include recessed portions that facilitate stretching.
 17. The assembly of claim 16, wherein the additional module includes an additional module projection; the second additional module includes a second additional module recess; and a depth of the recessed portions corresponds to an offset of the additional module projection and the second additional module recess.
 18. The assembly of claim 12, wherein: the second additional module includes a second additional module housing; the second additional module housing includes a first sidewall and a second sidewall; and the first sidewall and the second sidewall of the second additional module housing include tapered edges.
 19. The assembly of claim 12, wherein the first end module is disposed partially in the additional module; the additional module is disposed partially in the second additional module; and the second additional module is partially disposed in a third additional module or the second end module.
 20. A vehicle seat, comprising: the assembly of claim 1; and one or more fluid bladders in fluid communication with the assembly.
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